Course Handout

Institute/School Name	Chitkara University Institute of Engineering and Technology		
Department Name	Department of Computer Science & Engineering		
Programme Name	Bachelor of Engineering (B.E.), Computer Science & Engineering		
Course Name	Discrete Structures	Session	Jan – June 2026
Course Code	24APS0105	Semester/Batch	4th /2024
L-T-P(Per Week)	3-0-0	Course Credits	03
Pre-requisite	12th Level Mathematics, Combinatorics	NHEQF Level[1]	5
Course Coordinator	Dr. Mohit Dhuria	SDG Number[4]	1,2,9

1. Objectives of the Course

The course provides a wide scope of learning & understanding of the subject and the main objectives of the course are:

- To understand the theory and techniques of logic, graphs and trees, and algebraic systems.
- To apply the knowledge and skills obtained to investigate and solve a variety of discrete mathematical problems.
- To communicate mathematical ideas and to make effective use of appropriate technology.
- To comprehend discrete structures and their relevance within the context of computer science, in the areas of data structures and algorithms, in particular (Enabling Knowledge).
- To apply discrete structures into other computing problems such as formal specification, verification, databases, and cryptography (Problem Solving).

2. Course Learning Outcomes (CLOs)

Student should be able to:

	CLOs	Program Outcomes (PO)	NHEQF Level Descriptor[2]	No. of Lectures
CLO01	To apply the knowledge obtained to investigate and solve a variety of live problems.	PO1, PO2, PO11, PSO1.	Q1,Q2,Q3	**15**
CLO02	Analysis of relevance within the context of computer science, in the areas of data structures and algorithms	PO1, PO2, PSO1.	Q3,Q4	**15**
CLO03	To understand and apply the theory and techniques of Lattice, Logic and Boolean algebra.	PO2, PO3, PSO2, PSO3.	Q3, Q4,Q5	**10**
CLO04	To comprehend Graph Theory and its relevance within the context of computer science and finding solutions of live problems related to shortest path etc.	PO1, PO2, P03, PSO2, PSO3.	Q1,Q2,Q3,Q5	**5**
Total Contact Hours				**45**

CLO-PO Mapping

CLO	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12	PSO1	PSO2	Type of Assessment's[3]
CLO01	H	H									H		H		Summative assessments
CLO02	H	H												H	Summative assessments
CLO03		H	M										M		Summative assessments
CLO04	H	M	H											H	Summative assessments

H=High, M=Medium, L=Low

3. Recommended Books:

B01: Elements of Discrete Mathematics, by C.L.Liu, McGraw-Hill.
B02: Discrete Mathematics by Babu Ram, Pearson Education India.
B03: Discrete Mathematics by Lipschutz/Lipson, Schaum Series, 2nd Edition, TMH, Edition.
B04: Discrete Mathematics and its Applications by Kenneth H. Rosen, McGraw- Hill.
B05: Logic and Discrete Mathematics by Trembly/Grassmann, Pearson Education.
B06: Discrete Mathematics, Chitkara University Publications.
E-Resources: https://library.chitkara.edu.in/subscribed-books.php

4. Other readings and relevant websites:

[1] National Higher Education Qualification Framework Level, Refer to annexure
[2] NHEQF Level Descriptor, Refer to Annexure & Learning outcomes descriptors for qualification for all levels on the NHEQF
[3]Types of Assessments can be referred from Type of Assessments. Refer to Annexure.
[4]For SDG Mapping with Courses, Pl refer SDG Mapping policy for Courses

Serial No	Link of Journals, Magazines, websites and Research Papers
1.	http://download.nos.org/srsec311new/L.No.15-A.pdf
2.	http://faculty.simpson.edu/lydia.sinapova/www/cmsc180/LN180_Johnsonbaugh-07/L17-EquivalenceRel.htm
3.	http://web.cs.wpi.edu/~cs504/s00m/notes/recurrence/solve/step2/step2.html
4.	http://www.iep.utm.edu/prop-log/
5.	http://plato.stanford.edu/entries/boolalg-math/
6.	http://www.hamilton.ie/ollie/Downloads/Graph.pdf

5. Recommended Tools and Platforms

MATLAB, Wolfram Mathematica, Python and Sage.

6. Course Plan: Theory Plan

Lect. No.	Topic(s)
1.	**Recurrence relation**: Homogeneous recurrence relation with constant coefficients.
2.	**Recurrence relation**: Non-homogeneous recurrence relation with constant coefficients.
3.	**Characteristic polynomial & Introduction to generating functions:** Methods of generating functions
4.	**Characteristic polynomial & Introduction to generating functions:** Methods of generating functions
5.	**Characteristic polynomial & Introduction to generating functions:** Methods of generating functions
6.	**Logic**: Propositions & logical operators.
7.	**Logic**: Truth table. Laws of logic & Quantifiers,
8.	**Logic**: Validity of arguments, Truth values
9.	**Lattices**: Definition of lattice, Sub lattice, Isomorphic lattices,
10.	**Lattices**: Bounded lattice, Distributive lattice,
11.	**Lattices**: Modular lattice.
12.	**Lattices**: Modular lattice.
13.	**Boolean Algebra**: Atoms, Boolean expressions,
14.	**Boolean Algebra**: Applications of Boolean algebra to switching theory
15.	**Boolean Algebra**: Logic Gates.
Sessional Test-1 (1– 15 Lectures)	
16.	**Graph Theory:** Introduction to graphs & Sub graphs. Directed & Undirected graph,
17.	**Graph Theory:** Order& Size of graph, Degree of vertex,
18.	**Graph Theory**: Source, sink, Eccentricity, Trail, Walk, Path,
19.	**Graph Theory:** Distance, Diameter, Cycle, Wheel.
20.	**Graph Theory:** Multi graph, Planer graph,
21.	**Graph Theory:** Pseudo graph, weighted graph,
22.	**Graph Theory:** Regular, Complete
23.	**Graph Theory:** Traversable graph.
24.	**Graph Theory:** Isomorphism of graphs,
25.	**Graph Theory:** Sub graphs, Complement of graph,
26.	**Graph Theory:** Adjacency Matrix, Adjacency List, Incidence Matrix.
27.	**Graph Theory:** Complete Bipartite Graph & Spanning Graph,
28.	**Graph Theory:** Cut vertex, Cut edge,
29.	**Graph Theory:** Region, Eulerian & Hamiltonian graphs
30.	**Graph Theory:** Hamiltonian graphs
31.	**Euler's formula & its applications:** Traveling salesman problem.
32.	**Euler's formula & its applications:** Konigsberg Bridge problem.
33.	**Euler's formula & its applications** : Chromatic number & Graph coloring by Welsh Powell Algorithm.
34.	**Euler's formula & its applications**: Chromatic number & Graph coloring by Welsh Powell Algorithm.
Sessional Test-1I (1– 34 Lectures)	
35.	**Tree**: Binary trees,
36.	**Tree**: Traversing binary trees
37.	**Tree**: Rooted & spanning tree**,**
38.	**Tree**: Algebraic expression trees.
39.	**Tree**: Depth-First Search Algorithm,
40.	**Tree**: Breadth-First Search Algorithm.
41.	**Shortest path algorithm:** Kruskal's Algorithm,.
42.	**Shortest path algorithm:** Dijkstra's Algorithm,.
43.	**Shortest path algorithm:** Wars hall's Algorithm
44.	**Shortest path algorithm:** Prim's Algorithm.
45.	**Shortest path algorithm:** Prim's Algorithm.
End Term Exam. (1– 45 Lectures)	

7. Delivery/Instructional Resources

Theory Plan:

Lect. No.	Topics	CLO	Book No, CH No, Page No	TLM[3]	ALM[4]	Web References	Audio-Video
1.	**Recurrence relation**: Homogeneous recurrence relation with constant coefficients.	CLO1	B01, B02	Lecture Discussion	Quiz/Test Questions	https://en.wikipedia.org/wiki/Discrete_mathematics https://www.cs.cornell.edu/~rafael/discmath.pdf	https://nptel.ac.in/courses/106/106/106106183/
2.	**Recurrence relation**: Non-homogeneous recurrence relation with constant coefficients.	CLO1	B01, B02	Lecture Discussion	Quiz/Test Questions	https://en.wikipedia.org/wiki/Discrete_mathematics https://www.cs.cornell.edu/~rafael/discmath.pdf	https://nptel.ac.in/courses/106/106/106106183/
3.	**Characteristic polynomial & Introduction to generating functions:** Methods of generating functions	CLO1	B01, B02	Lecture Discussion	Quiz/Test Questions	https://web.stanford.edu/class/cs103x/cs103x-notes.pdf https://en.wikipedia.org/wiki/Discrete_mat	https://nptel.ac.in/courses/106/106/106106183/
4.	**Characteristic polynomial & Introduction to generating functions:** Methods of generating functions	CLO1	B01, B02	Lecture Discussion	Quiz/Test Questions	https://web.stanford.edu/class/cs103x/cs103x-notes.pdf https://en.wikipedia.org/wiki/Discrete_mat	https://nptel.ac.in/courses/106/106/106106183/
5.	**Characteristic polynomial & Introduction to generating functions:** Methods of generating functions	CLO1	B01, B02	Lecture Discussion	Quiz/Test Questions	https://web.stanford.edu/class/cs103x/cs103x-notes.pdf https://en.wikipedia.org/wiki/Discrete_mat	https://nptel.ac.in/courses/106/106/106106183/
6.	**Logic**: Propositions & logical operators.	CLO1	B01, B06	Lecture Discussion	Quiz/Test Questions	https://en.wikipedia.org/wiki/Discrete_mathematics https://www.cs.cornell.edu/~rafael/discmath.pdf	https://nptel.ac.in/courses/106/106/106106183/
7.	**Logic**: Truth table. Laws of logic & Quantifiers,	CLO1	B01, B06	Lecture and Discussion	Quiz/Test Questions	https://en.wikipedia.org/wiki/Discrete_mathematics https://www.cs.cornell.edu/~rafael/discmath.pdf	https://nptel.ac.in/courses/106/106/106106183/
8.	**Logic**: Validity of arguments, Truth values	CLO1	B01, B06	Lecture Discussion	Quiz/Test Questions	https://en.wikipedia.org/wiki/Discrete_mathematics https://www.cs.cornell.edu/~rafael/discmath.pdf	https://nptel.ac.in/courses/106/106/106106183/
9.	**Lattices**: Definition of lattice, Sub lattice, Isomorphic lattices,	CLO1	B01, B02	Lecture Discussion	Quiz/Test Questions	https://web.stanford.edu/class/cs103x/cs103x-notes.pdf https://en.wikipedia.org/wiki/Discrete_mat	https://nptel.ac.in/courses/106/106/106106183/
10.	**Lattices**: Bounded lattice, Distributive lattice,	CLO1	B01, B02	Lecture Discussion	Quiz/Test Questions	https://web.stanford.edu/class/cs103x/cs103x-notes.pdf https://en.wikipedia.org/wiki/Discrete_mat	https://nptel.ac.in/courses/106/106/106106183/
11.	**Lattices**: Modular lattice.	CLO1	B01, B02	Lecture Discussion	Quiz/Test Questions	https://web.stanford.edu/class/cs103x/cs103x-notes.pdf https://en.wikipedia.org/wiki/Discrete_mat	https://nptel.ac.in/courses/106/106/106106183/
12.	**Lattices**: Modular lattice.	CLO1	B01, B02	Lecture Discussion	Quiz/Test Questions	https://web.stanford.edu/class/cs103x/cs103x-notes.pdf https://en.wikipedia.org/wiki/Discrete_mat	https://nptel.ac.in/courses/106/106/106106183/
13.	**Boolean Algebra**: Atoms, Boolean expressions,	CLO1	B01, B06	Lecture Discussion	Quiz/ Test Questions	https://web.stanford.edu/class/cs103x/cs103x-notes.pdf https://en.wikipedia.org/wiki/Discrete_mat	https://nptel.ac.in/courses/106/106/106106183/
14.	**Boolean Algebra**: Applications of Boolean algebra to switching theory	CLO1	B01, B06	Lecture Discussion	Quiz/Test Questions	https://web.stanford.edu/class/cs103x/cs103x-notes.pdf https://en.wikipedia.org/wiki/Discrete_mat	https://nptel.ac.in/courses/106/106/106106183/
15.	**Boolean Algebra**: Logic Gates.	CLO1	B01, B06	Lecture Discussion	Quiz/Test Questions	https://web.stanford.edu/class/cs103x/cs103x-notes.pdfhttps://en.wikipedia.org/wiki/Discrete_mat	https://nptel.ac.in/courses/106/106/106106183/

[3] Teaching Learning Methods
[4] Active Learning Methods

16.	**Graph Theory:** Introduction to graphs & Sub graphs. Directed & Undirected graph,	CLO2	B02	Lecture Discussion	Quiz/Test Questions	https://en.wikipedia.org/wiki/Discrete_mathematics https://www.cs.cornell.edu/~rafael/discmath.pdf	https://nptel.ac.in/courses/106/106/106106183/
17.	**Graph Theory:** Order& Size of graph, Degree of vertex,	CLO2	B02	Lecture Discussion	Quiz/Test Questions	https://en.wikipedia.org/wiki/Discrete_mathematics https://www.cs.cornell.edu/~rafael/discmath.pdf	https://nptel.ac.in/courses/106/106/106106183/
18.	**Graph Theory**: Source, sink, Eccentricity, Trail, Walk, Path,	CLO2	B02	Lecture	Quiz/Test Questions	https://en.wikipedia.org/wiki/Discrete_mathematics https://www.cs.cornell.edu/~rafael/discmath.pdf	https://nptel.ac.in/courses/106/106/106106183/
19.	**Graph Theory:** Distance, Diameter, Cycle, Wheel.	CLO2	B02	Lecture Discussion	Quiz/Test Questions	https://en.wikipedia.org/wiki/Discrete_mathematics https://www.cs.cornell.edu/~rafael/discmath.pdf	https://nptel.ac.in/courses/106/106/106106183/
20.	**Graph Theory:** Multi graph, Planer graph,	CLO2	B06	Lecture Discussion	Quiz/Test Questions	https://web.stanford.edu/class/cs103x/cs103x-notes.pdf https://en.wikipedia.org/wiki/Discrete_mat	https://nptel.ac.in/courses/106/106/106106183/
21.	**Graph Theory:** Pseudo graph, weighted graph,	CLO2	B06	Lecture Discussion	Quiz/Test Questions	https://web.stanford.edu/class/cs103x/cs103x-notes.pdf https://en.wikipedia.org/wiki/Discrete_mat	https://nptel.ac.in/courses/106/106/106106183/
22.	**Graph Theory:** Regular, Complete	CLO2	B06	Lecture Discussion	Quiz/Test Questions	https://web.stanford.edu/class/cs103x/cs103x-notes.pdf https://en.wikipedia.org/wiki/Discrete_mat	https://nptel.ac.in/courses/106/106/106106183/
23.	**Graph Theory:** Traversable graph.	CLO2	B06	Lecture Discussion	Quiz/Test Questions	https://web.stanford.edu/class/cs103x/cs103x-notes.pdf https://en.wikipedia.org/wiki/Discrete_mat	https://nptel.ac.in/courses/106/106/106106183/
24.	**Graph Theory:** Isomorphism of graphs,	CLO2	B06	Lecture Discussion	Quiz/Test Questions	https://web.stanford.edu/class/cs103x/cs103x-notes.pdf https://en.wikipedia.org/wiki/Discrete_mat	https://nptel.ac.in/courses/106/106/106106183/
25.	**Graph Theory:** Sub graphs, Complement of graph,	CLO2	B06	Lecture Discussion	Quiz/Test Questions	https://en.wikipedia.org/wiki/Discrete_mathematics https://www.cs.cornell.edu/~rafael/discmath.pdf	https://nptel.ac.in/courses/106/106/106106183/
26.	**Graph Theory:** Adjacency Matrix, Adjacency List, Incidence Matrix.	CLO2	B06	Lecture Discussion	Quiz/Test Questions	https://en.wikipedia.org/wiki/Discrete_mathematics https://www.cs.cornell.edu/~rafael/discmath.pdf	https://nptel.ac.in/courses/106/106/106106183/
27.	**Graph Theory:** Complete Bipartite Graph & Spanning Graph,	CLO2	B03	Lecture Discussion	Quiz/Test Questions	https://web.stanford.edu/class/cs103x/cs103x-notes.pdf https://en.wikipedia.org/wiki/Discrete_mat	https://nptel.ac.in/courses/106/106/106106183/
28.	**Graph Theory:** Cut vertex, Cut edge,	CLO2	B03	Lecture Discussion	Quiz/Test Questions	https://web.stanford.edu/class/cs103x/cs103x-notes.pdf https://en.wikipedia.org/wiki/Discrete_mat	https://nptel.ac.in/courses/106/106/106106183/
29.	**Graph Theory:** Region, Eulerian & Hamiltonian graphs	CLO2	B03	Lecture Discussion	Quiz/Test Questions	https://web.stanford.edu/class/cs103x/cs103x-notes.pdf https://en.wikipedia.org/wiki/Discrete_mat	https://nptel.ac.in/courses/106/106/106106183/
30.	**Graph Theory:** Hamiltonian graphs	CLO2	B03	Lecture Discussion	Quiz/Test Questions	https://web.stanford.edu/class/cs103x/cs103x-notes.pdf https://en.wikipedia.org/wiki/Discrete_mat	https://nptel.ac.in/courses/106/106/106106183/
31.	**Euler's formula & its applications:** Traveling salesman problem.	CLO3	B02	Lecture Discussion	Quiz/Test Questions	https://en.wikipedia.org/wiki/Discrete_mathematics https://www.cs.cornell.edu/~rafael/discmath.pdf	https://nptel.ac.in/courses/106/106/106106183/
32.	**Euler's formula & its applications:** Konigsberg Bridge problem.	CLO3	B02	Lecture Discussion	Quiz/Test Questions	https://en.wikipedia.org/wiki/Discrete_mathematics https://www.cs.cornell.edu/~rafael/discmath.pdf	https://nptel.ac.in/courses/106/106/106106183/
33.	**Euler's formula & its applications** : Chromatic number &	CLO3	B02	Lecture Discussion	Quiz/Test Questions	https://web.stanford.edu/class/cs103x/cs103x-notes.pdf https://en.wikipedia.org/wiki/Discr	https://nptel.ac.in/courses/106/106/106106183/

	Graph coloring by Welsh Powell Algorithm.					ete_mat	
34.	**Euler's formula & its applications**: Chromatic number & Graph coloring by Welsh Powell Algorithm.	CLO3	B02	Lecture Discussion	Quiz/Test Questions	https://web.stanford.edu/class/cs103x/cs103x-notes.pdf https://en.wikipedia.org/wiki/Discrete_mat	https://nptel.ac.in/courses/106/106/106106183/
35.	**Tree**: Binary trees,	CLO4	B06	Lecture Discussion	Quiz/Test Questions	https://en.wikipedia.org/wiki/Discrete_mathematics https://www.cs.cornell.edu/~rafael/discmath.pdf	https://nptel.ac.in/courses/106/106/106106183/
36.	**Tree**: Traversing binary trees	CLO4	B06	Lecture Discussion	Quiz/Test Questions	https://en.wikipedia.org/wiki/Discrete_mathematics https://www.cs.cornell.edu/~rafael/discmath.pdf	https://nptel.ac.in/courses/106/106/106106183/
37.	**Tree**: Rooted & spanning tree**,**	CLO4	B06	Lecture Discussion	Quiz/Test Questions	https://web.stanford.edu/class/cs103x/cs103x-notes.pdf https://en.wikipedia.org/wiki/Discrete_mat	https://nptel.ac.in/courses/106/106/106106183/
38.	**Tree**: Algebraic expression trees.	CLO4	B06	Lecture Discussion	Quiz/Test Questions	https://web.stanford.edu/class/cs103x/cs103x-notes.pdf https://en.wikipedia.org/wiki/Discrete_mat	https://nptel.ac.in/courses/106/106/106106183/
39.	**Tree**: Depth-First Search Algorithm,	CLO4	B06	Lecture Discussion	Quiz/Test Questions	https://en.wikipedia.org/wiki/Discrete_mathematics https://www.cs.cornell.edu/~rafael/discmath.pdf	https://nptel.ac.in/courses/106/106/106106183/
40.	**Tree**: Breadth-First Search Algorithm.	CLO4	B06	Lecture Discussion	Quiz/Test Questions	https://en.wikipedia.org/wiki/Discrete_mathematics https://www.cs.cornell.edu/~rafael/discmath.pdf	https://nptel.ac.in/courses/106/106/106106183/
41.	**Shortest path algorithm:** Kruskal's Algorithm,.	CLO4	B05	Lecture Discussion	Quiz/Test Questions	https://web.stanford.edu/class/cs103x/cs103x-notes.pdf	https://nptel.ac.in/courses/106/106/106106183/
42.	**Shortest path algorithm:** Dijkstra's Algorithm,.	CLO4	B05	Lecture Discussion	Quiz/Test Questions	https://web.stanford.edu/class/cs103x/cs103x-notes.pdf	
43.	**Shortest path algorithm:** Wars hall's Algorithm	CLO4	B05	Lecture Discussion	Quiz/Test Questions	https://en.wikipedia.org/wiki/Discrete_mat,	https://nptel.ac.in/courses/106/106/106106183/
44.	**Shortest path algorithm:** Prim's Algorithm.	CLO4	B05	Lecture Discussion	Quiz/Test Questions	https://en.wikipedia.org/wiki/Discrete_mat,	https://nptel.ac.in/courses/106/106/106106183/
45.	**Shortest path algorithm:** Prim's Algorithm.	CLO4	B05	Lecture Discussion	Quiz/Test Questions	https://en.wikipedia.org/wiki/Discrete_mathematics https://www.cs.cornell.edu/~rafael/discmath.pdf	https://nptel.ac.in/courses/106/106/106106183/

8. Remedial Classes[5]

After Every Sessional Test, weak learners will be identified, and supplement course handout will be provided. Student list and Impact Observed report will be submitted to Dean through proper channel.

9. Self-Learning[6]

Assignments to promote self-learning, survey of contents from multiple sources.

S. No	Topics	CLO	ALM	References/MOOCS
1.	To solve a variety of live problems related	CLO1	Assignment	Reference Books
2.	Analysis of relevance within the context of computer science, in the areas of data structures and algorithms	CLO2	Assignment	Reference Books
3.	To understand and apply the theory and techniques of Lattice, Logic and Boolean algebra.	CLO3	Assignment	Reference Books
4.	To comprehend Graph Theory and its relevance within the context of computer science and finding solutions of live problems related to shortest path etc.	CLO4	Assignment	Reference Books

[5] Refer to Annexure
[6] Refer to Annexure

10. Delivery Details of Content Beyond Syllabus[7]

Content beyond the syllabus covered (if any) should be delivered to all students that would be planned, and schedule notified accordingly.

S.No	Advanced Topics, Additional Reading, Research papers and any	CLO	POs	ALM	References/MOOCS
NA	NA	NA	NA	NA	NA

11. Evaluation Scheme & Components:

Assessment Type[8]	Evaluation Component[9]	Type of Component[10]	No. of Assessments[11]	% Weightage of Component	Max. Marks	Mode of Assessment	CLO
Summative	Component1	Sessional Tests (STs)	02*	50%	**50**	Online/ Offline	CLO01, CLO02, CLO03.
Summative	Component2	End Term Examination	01**	50%	**50**	Online/ Offline	CLO01, CLO02, CLO03, CLO04,
	Total		**100%**				

*Students will have to appear in all Sessional Tests.
** As per Academic Guidelines, a minimum of 75% attendance is required to become eligible for appearing in the End Semester Examination.

12. Syllabus of the Course:

Subject: Discrete Structures, 22APS0105			
S. No.	**Topic(s)**	**No. of Lectures**	**Weightage %**
1	**Recurrence relation**: Homogeneous and Non-homogeneous recurrence relation with constant coefficients, **Characteristic polynomial & Introduction to generating functions:** Methods of generating functions.	**5**	**10**
2	**Logic**: Propositions & logical operators, Truth table. Laws of logic & Quantifiers, Validity of arguments, Truth values. **Lattices**: Definition of lattice, Sub lattice, Isomorphic lattices, Bounded lattice, Distributive lattice	**5**	**10**
3	**Lattices**:, complemented lattice, Modular lattice. **Boolean Algebra**: Atoms, Boolean Sex pressions, Applications of Boolean algebra to switching theory & Logic Gates.	**5**	**10**
Sessional Test 1 (ST1)			
4	**Graph Theory:** Introduction to graphs & Sub graphs. Directed & Undirected graph, Order& Size of graph, Degree of vertex, Source, sink, Eccentricity, Trail, Walk, Path, Distance, Diameter, Cycle, Wheel., Multi graph, Planer graph, Pseudo graph, weighted graph, Regular, Complete and Traversable graph. Isomorphism of graphs, Sub graphs, complement of graph, Adjacency Matrix, Adjacency List, Incidence Matrix, Complete Bipartite Graph & Spanning Graph, cut vertex, Cut edge, Region, Eulerian & Hamiltonian graphs.	**15**	**35**
5	**Euler's formula & its applications:** Traveling salesman problem, Konigsberg Bridge problem, Chromatic number & Graph coloring by Welsh Powell Algorithm.	**4**	**15**
Sessional Test 2 (ST2) (ST1 syllabus also included)			
6	**Tree**: Binary trees, traversing binary trees, Rooted & spanning tree, Algebraic expression trees, Depth-First Search Algorithm, Breadth-First Search Algorithm. **Shortest path Algorithm:** Kruskal's Algorithm, Dijkstra's Algorithm, Warshall's Algorithm, Prim's Algorithm.	**11**	**20**
End Term Exam (ETE)			

13. Academic Integrity Policy:

Education at Chitkara University builds on the principle that excellence requires freedom where Honesty and integrity are its prerequisites. Academic honesty in the advancement of knowledge requires that all students and Faculty respect the integrity of one another's work and recognize the importance of acknowledging and safeguarding intellectual property. Any breach of the same will be tantamount to severe academic penalties.

[7] Refer to Annexure
[8] Refer to Annexure 2 of NCrF
[9] Refer to Annexure
[10] Refer to Annexure
[11] Refer to Annexure

This Document is approved by:

Designation	Name	Signature
Course Coordinator	Dr. Mohit Dhuria	
Program Head	Prof. Reetu Malhotra	
Dean	Prof. Mohit Kumar Kakkar	
Date (DD/MM/YYYY)	16.01.2026	

Annexure

1. **Pre- requisite**
   Mention The Pre-requisite skill set or course/s if it is expected to be studies before this course, otherwise write "not applicable".

2. **NHEQF levels**

The NHEQF levels represent a series of sequential stages expressed in terms of a range of learning outcomes against which typical qualifications are positioned/located. NHEQF level 4.5 represents learning outcomes appropriate to the first year (first two semesters) of the undergraduate programme of study, while Level 8 represents learning outcomes appropriate to the doctoral-level programme of study.
Table 1: Higher education qualifications at different levels on the NHEQF

NHEQF level	Examples of higher education qualifications located within each level
Level 4.5	Undergraduate Certificate. Programme duration: First year (first two semesters) of theundergraduate programme, followed by an exit 4-credit skills-enhancement course(s).
Level 5	Undergraduate Diploma. Programme duration: First two years (first four semesters) of the undergraduate programme, followed by an exit 4-credit skills-enhancement course(s) lasting two months.
Level 5.5	Bachelor's Degree. Programme duration: First three years (Six semesters) of the four-yearundergraduate programme.
Level 6	Bachelor's Degree (Honours/ Honours with Research). Programme duration: Four years(eight semesters).
Level 6	Post-Graduate Diploma. Programme duration: One year (two semesters) for those whoexit after successful completion of the first year (two semesters) of the 2-year master's programme.
Level 6.5	Master's degree. (e.g. M.A., M.Com., M.Sc., etc.) Programme duration: Two years (four semesters) after obtaining a 3- year Bachelor's degree (e.g. B.A., B.Sc., B.Com.etc.).
Level 6.5	Master's degree. (e.g. M.A., M.Com., M.Sc., etc.) Programme duration: One year (two semesters) after obtaining a 4 - year Bachelor's degree (Honours/ Honours with Research) (e.g. B.A., B.Sc., B.Com. etc.).
Level 7	Master's degree.(e.g. M.E./M.Tech. etc.) Programme duration: Two years (four semesters) after obtaining a 4-year Bachelor's degree. (e.g. B.E./B.Tech. etc.)
Level 8	Doctoral Degree

3. **NHEQF level descriptors**

Each NHEQF level is structured based on the defined learning outcomes which lead to the expected graduate attributes/profile. The level descriptors reflect the expected outcomes of learning that should be achieved and demonstrated by graduates of a specific programme of study leading to a qualification at a specific NHEQF level.

Click Learning outcomes descriptors for qualification for all levels on the NHEQF

4. **Course Outcomes**
   The number of Course Outcomes is recommended to be 4-5 for courses that do not contain practical component and 6 for those courses with a practical component. Flexibility can be sought by the post-graduate courses in this regard.

5. **Theory/lab Plan**
   The following are the guidelines to be followed while creating plans

- Each session may be planned for a duration of 45/50mins (irrespective of the double hour or single hour scheduled in timetable).
- Every session must incorporate at least one active learning method which may or may not be part of the assessments.
- Put BoS Approved Syllabus in the topics. Deviations (if any) from BoS approved syllabus must be brought to the notice of BoS chairman & Dean Academics, After approval, revised handout should be submitted.
- The Topics elaborated in the Theory/Lab plan must match those in the course execution plan.

6. **Teaching Learning Methods**
   The following are some of the Teaching & Learning methods that can be incorporated in session wise teaching learning plan.

- **Teacher-centered Learning Methods:**
  i. Lecture
  ii. Discussion
  iii. Demonstration method using a simulation or a tool
  iv. Reviewing
  v. Questioning

- **Learner-centered teaching & Learning methods:**
  i. **Active learning**, in which students solve problems, answer questions, formulate questions of their own, discuss, explain, debate, or brainstorm during class;
  ii. **Cooperative learning**, in which students work in teams on problems and projects under conditions that assure both positive interdependence and individual accountability; and
  iii. **Inductive teaching and learning**, in which students are first presented with challenges (questions or problems) and learn the course material in the context of addressing the challenges.
  iv. **Inductive methods** include inquiry-based learning, case based instruction, problem-based learning, project-based learning, discovery learning, and just-in-time teaching. It is important to integrate authentic, reflective and collaborative learning experiences when designing for student-centered learning.

7. **Active Learning Methods**

The following are some of the Active Learning Methods that can be incorporated in session wise teaching learning plan.

- One Minute Paper
- Group Discussion
- Student-Created PPT, Charts, Matrices, Flowcharts, Models
- The Fish Bowl
- Debate
- Video Synthesis
- Quiz/Test Questions
- Brain Storming Sessions
- Case Study
- Shadowing
- Leading Question
- Puzzle, Enigma, Contradiction
- Statement-Opinion-Summary
- Think / Pair / Share
- Peer Review
- Just in Time Teaching
- Statement-Opinion-Summary
- Peer Survey
- Focused Listing
- Role-Playing
- Student Field Work with Reflection
- Infusing Humor into Class Sessions
- Inviting Effective Guest Speakers

8. Remedial Classes

After every Sessional Test, identify weak learners, provide supplement course handout. Student list and Impact Observed report should be submitted to Dean through proper channel.

9. Self Learning

Plan 10% of topics in self-learning mode with discussions, ALM's and Assessment happing in the class.

10. Content Beyond Syllabus

Plan Advanced Topics, Experiments, Additional Reading, Research papers in self-learning mode with ALM's and Assessment happing in the regular class or lab. Usually caters advanced learners. Identify Advanced learners. For Extra classes, schedule should be notified accordingly.

11. Assessment Type

1. Assessment broadly can be classified into the following types:

a. **Diagnostic assessments**: Diagnostic assessments are intended to help teachers identify what students know and can do in different domains to support their students' learning. These help teachers determine strengths of students in various areas to
better address their specific needs.
b. **Formative assessments**: Formative assessment refers to a wide variety of methods that teachers use to conduct in-process evaluations of student comprehension, learning needs, and academic progress during a lesson, unit, or a course. Formative assessments help teachers identify concepts that students are struggling to understand, skills they are having difficulty acquiring, or learning standards they have not yet achieved so that adjustments can be made to lessons, instructional techniques, and academic support.
c. **Summative assessments**: Summative assessment is an assessment administered at the end of an instructional unit in a course. These assessments are intended to evaluate student learning by comparing performance to a standard or benchmark.
d. **Ipsative assessments**: Ipsative assessment involves comparisons between past and current work to identify a learner's growth over time, rather than progress toward an external set of criteria. Therefore, Ipsative assessment is an internal or self-referenced assessment.
e. **Norm-referenced assessments**: Norm-referenced tests report whether test takers performed better or worse than a hypothetical average student, which is determined by comparing scores against the performance results of a statistically selected group of test takers, typically of the same age or grade level, who have already taken the exam.
f. **Criterion-referenced assessments**: Criterion-Reference tests measure the performance of test takers against the criteria covered in the curriculum.

g. **Peer-to-Peer randomised Assessments**: Peers will be able to provide assessment in this case
h. **Industry Validation of Effectiveness**: In the Vocation Education, Industry validation of effectiveness of training is particularly important.
i. **Self-assessments**: To evaluate how much the learner has grasped by self-learning.

2. Other Assessment Methods: Conducting an assessment takes time, thought, attention, planning, and often collaboration. Each assessment tool, whether a short survey or detailed rubric, will be useful only insofar as it both addresses the outcomes well and is feasible to use.

a**. Rubrics**: For assessing qualitative student work such as essays, projects, reports, or presentations. Rubrics serve well to clearly denote the specific expectations for an assignment, for collecting data for assessment of student learning outcomes. and for student performance. Rubrics can be used for grading, for providing feedback to students, and for informing and encouraging students to think about their own learning.
b. **Portfolios and E-Portfolio**: Portfolios can provide a window into the process of student learning across a semester-long project that can be assessed (usually by using a rubric).
c. **Curriculum Mapping**: A good curriculum map can serve to focus assessment, and the improvements that follow, where it will be most useful, informative, or effective.
d. **Structured Interviews**: While time-consuming, structured interviews are useful when specific questions need to be asked. It also leaves room for unplanned topics or ideas to emerge.
e. **Student Experience Surveys**: Student experience in research universities (SERU), including administration of on-line census SERU Undergraduate and Graduate Surveys, can yield important information about student perceptions and experiences.

12. Evaluation Component & Types

As per LMs we need to figure it out whether it is component 1, 2 or 3. In Types of Evaluation Component, we need to specify what type of evaluation we are performing like Continuous Evaluation or Sessional Test or End Term Examination.

13. No. of Assessments and Weightage of Components

Department will give guideline for number of assessments, mandatory or optional and weightage.